

PROGRAM:

CB3491_Cyber_Security / Cyber_Security_814723104109.ipynb

↑ To

Preview

Code

Blame

18261 lines (18261 loc) · 1010 KB



Raw



In [10]:

```
# 🌐 Visualizing Global Cybersecurity Threats (2015–2024)
# Author: Nowshika Mirza R
# Project: Mini Project – Cyber Threat Visualization

# =====
# Step 1: Import Libraries
# =====
import pandas as pd
import plotly.express as px
import matplotlib.pyplot as plt
from wordcloud import WordCloud

# =====
# Step 2: Load Dataset
# =====
df = pd.read_csv('Global_Cybersecurity_Threats_2015–2024.csv')

# Quick look at the data
print("🔍 Dataset Overview:")
print(df.head(), "\n")
print("📊 Columns:", df.columns.tolist())

# =====
# Step 3: Clean & Prepare Data
# =====
# Convert 'Year' to string for categorical plotting
df['Year'] = df['Year'].astype(str)

# Fill missing values for visualization
df['Attack Type'] = df['Attack Type'].fillna('Unknown')
df['Severity'] = df.get('Severity', pd.Series()).fillna('Unknown') # If 'Severity' exists
df['Target Country'] = df.get('Target Country', pd.Series()).fillna('Unknown') # If exists

# =====
# Step 4: Financial Loss Over Years by Attack Type (Interactive)
# =====
fig1 = px.bar(
    df.groupby(['Year', 'Attack Type'])['Financial Loss (in Million $)'].sum().reset_index(),
    x='Year', y='Financial Loss (in Million $)', color='Attack Type',
    title='Financial Loss Over Years by Attack Type',
    labels={'Financial Loss (in Million $)': 'Loss (Million $)', 'Year': 'Year'},
    barmode='group', height=450
)
fig1.update_layout(xaxis_tickangle=-45, title_x=0.5)
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    )
fig1.update_layout(xaxis_tickangle=-45, title_x=0.5)
fig1.show()

# =====
# Step 5: Number of Affected Users by Target Industry (Top 10)
# =====
industry_affected = df.groupby('Target Industry')['Number of Affected Users'].sum().nlargest(10).reset_index()
fig2 = px.bar(
    industry_affected,
    y='Target Industry', x='Number of Affected Users', orientation='h',
    title='Top 10 Target Industries by Affected Users',
    labels={'Number of Affected Users': 'Affected Users', 'Target Industry': 'Industry'},
    height=450
)
fig2.update_layout(yaxis={'categoryorder': 'total ascending'}, title_x=0.5)
fig2.show()

# =====
# Step 6: Average Incident Resolution Time by Attack Source
# =====
source_resolution = df.groupby('Attack Source')['Incident Resolution Time (in Hours)'].mean().reset_index()
fig3 = px.bar(
    source_resolution,
    x='Attack Source', y='Incident Resolution Time (in Hours)',
    title='Average Resolution Time by Attack Source',
    labels={'Incident Resolution Time (in Hours)': 'Resolution Time (Hours)', 'Attack Source': 'Source'},
    height=400
)
fig3.update_layout(title_x=0.5)
fig3.show()

# =====
# Step 7: Word Cloud of Attack Types
# =====
text = ' '.join(df['Attack Type'].astype(str))

plt.figure(figsize=(12, 8))
wordcloud = WordCloud(width=1200, height=800, background_color='white', colormap='inferno').generate(text)
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Common Cyber Attack Types', fontsize=18, fontweight='bold')
plt.show()

```

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```
print("✅ Visualization Completed – Your dashboard is ready!")
```

Dataset Overview:

	Country	Year	Attack Type	Target Industry \
0	China	2019	Phishing	Education
1	China	2019	Ransomware	Retail
2	India	2017	Man-in-the-Middle	IT
3	UK	2024	Ransomware	Telecommunications
4	Germany	2018	Man-in-the-Middle	IT

	Financial Loss (in Million \$)	Number of Affected Users	Attack Source \
0	80.53	773169	Hacker Group
1	62.19	295961	Hacker Group
2	38.65	605895	Hacker Group
3	41.44	659320	Nation-state
4	74.41	810682	Insider

	Security Vulnerability Type	Defense Mechanism Used \
0	Unpatched Software	VPN
1	Unpatched Software	Firewall
2	Weak Passwords	VPN
3	Social Engineering	AI-based Detection
4	Social Engineering	VPN

	Incident Resolution Time (in Hours)
0	63
1	71
2	20
3	7
4	68

Columns: ['Country', 'Year', 'Attack Type', 'Target Industry', 'Financial Loss (in Million \$)', 'Number of Affected Users', 'Attack Source', 'Security Vulnerability Type', 'Defense Mechanism Used', 'Incident Resolution Time (in Hours)']

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import plotly.express as px
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('Global_Cybersecurity_Threats_2015-2024.csv')

# Convert Year to string for categorical plotting
df['Year'] = df['Year'].astype(str)

# =====
# Advanced Plot 1: Heatmap of Attack Types vs Target Industries
# =====
pivot_data = df.pivot_table(index='Attack Type', columns='Target Industry', values='Financial Loss (in Million $)')
plt.figure(figsize=(16, 8))
sns.heatmap(pivot_data, cmap='YlOrRd', linewidths=0.5, linecolor='gray')
plt.title('Heatmap: Attack Types vs Target Industries (Financial Loss)', fontsize=16, fontweight='bold')
plt.xlabel('Target Industry')
plt.ylabel('Attack Type')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

# =====
# Advanced Plot 2: Yearly Trend for Ransomware Financial Loss
# =====
ransomware_df = df[df['Attack Type'] == 'Ransomware'].groupby('Year')['Financial Loss (in Million $)'].sum().reset_index()
fig = px.line(ransomware_df, x='Year', y='Financial Loss (in Million $)', title='Yearly Financial Loss from Ransomware')
fig.update_layout(title_x=0.5, yaxis_title='Loss (Million $)')
fig.show()

# =====
# Advanced Plot 3: Correlation Heatmap of Numeric Features
# =====
numeric_cols = ['Financial Loss (in Million $)', 'Number of Affected Users', 'Incident Resolution Time (in Hours)']
corr = df[numeric_cols].corr()
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.title('Correlation Matrix of Key Numeric Features', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()
```